

International Finance

Money and Money Creation/Multiplication

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Money: Motivation

- Most of the topics in international finance are related to money
 - Exchange rate is a relative price of two monies
 - Balance of payments records flows values expressed in money
 - CIP & UIP are based on interest rate differentials (where interest rates are closely linked to monetary policy)
- It is therefore logical to start with supply and demand for money

What is Money?

- Any commodity that is **widely accepted** in exchange and **has stable value**
- If a commodity is to be used as money:
 - It must be widely accepted for payments
 - It should be standardized and not easy to forge
 - Its supply should be limited
 - It should be divisible
 - It should be easy to carry
 - It should not deteriorate quickly
- Some forms of money in history:
 - Coins, banknotes, precious metals, strings of beads, seashells, large stones, cloth, cigarettes, whisky...

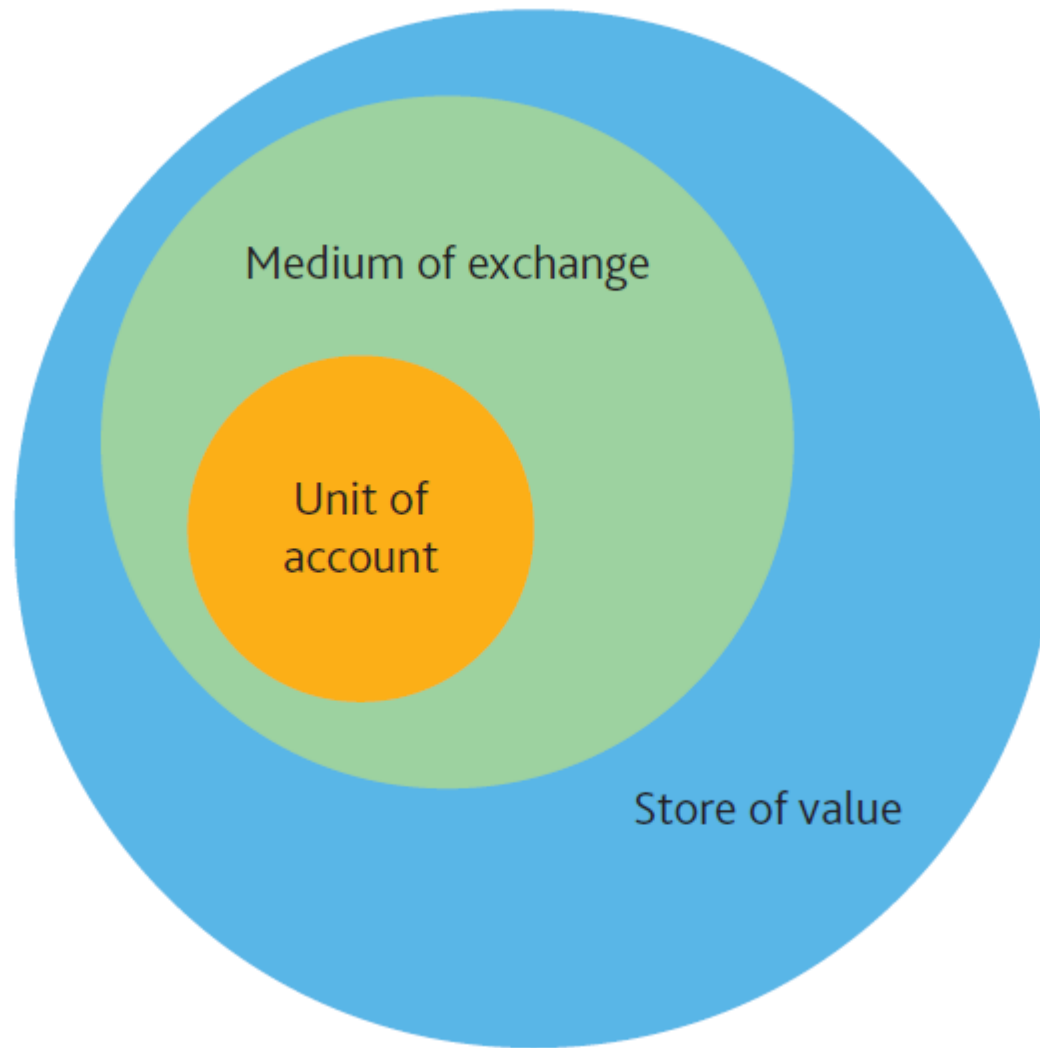
Stable Value?



Functions of Money

- Medium of exchange
- Unit of account
- Store of value

Hierarchy of the Functions?



Types of Money and Liquidity

- Commodity money;
 - Representative money;
 - Fiat money;
 - Credit money;
 - Digital currencies (Bitcoin and other)
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- **Liquidity** - easiness with which money is converted into other things, goods and services.

Money in International Transactions

- Types of currencies
 - National currencies (+ instruments based on them)
 - Artificial currencies (SDR)
 - Digital currencies/cryptocurrencies (Bitcoin, Litecoin, Peercoin...)
 - Commodities?
- Useability of domestic currency in international transactions
 - Role of convertibility
 - Non-convertible currencies
 - “Internal convertibility”
 - Convertibility on the current account
 - “Full convertibility”
 - Role of credibility, stability,
 - Vehicle currencies v. exotic currencies

Money Supply and Monetary Policy

- **Money supply** – is the total amount of money available in an economy at a particular point in time;
- The control over the money supply is called **Monetary Policy**;
- Monetary policy is conducted by a **Central Bank**:
 - In the US – the Federal Reserve, or the **Fed**;
 - In Eurozone – **European Central Bank (ECB)**

Money Stock Measures – FED I.

- **M1**

- + currency outside the U.S. Treasury, FED banks and the vaults of depository institutions
- + travelers checks of non-bank issuers
- + demand deposits at commercial banks
- + other checkable deposits

- **M2**

- + M1
- + savings deposits (incl. money market deposit accounts)
- + small-denomination time deposits (< 100 000 \$)
- + balances in retail money market mutual funds

Monetary Base

- = the part of money supply that CEB can influence “directly”
- How? Monetary base is in the balance sheet of the central bank, it can be influenced by the bank’s policy
- Total money supply is related to the monetary base by a multiplication process

CEB

Assets

FOREX reserves

Loan to banks

Govt. securities (e.g. T-bills)

....

Other Assets

Liabilities

Reserves of commercial banks

Currency in circulation

...

Other liabilities

CEB

Assets		Liabilities	
FOREX reserves	(+100)	Reserves of banking sector	+100
Loan to banks	(+100)	Currency in circulation	
Govt. securities	(+100)		
....		...	
Other Assets		Other liabilities	

Initiation of the multiplication process: the CEB buys from a commercial bank some foreign exchange or securities, alternatively it gives it a loan

Commercial Bank (1st)

Assets

Securities	-100
Reserves	+100
<i>Loans</i>	<i>+100</i>

Liabilities

<i>Checkable deposits</i>	<i>+100</i>
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Commercial Bank (1st)

Assets

Liabilities

Securities -100

Loans +100

The result: holdings of securities decreased, but the loans to firms increased.
But the process does not end here ...

Commercial Bank (A)

Assets

Liabilities

Reserves: + 10

Loans + 90

Deposits: + 100

Commercial Bank (B)

Assets

Liabilities

Reserves: + 9

Loans + 81

Deposits: + 90

Deposit Creation (10% RR)

Bank	Change in Deposits	Change in Loans	Change in Reserves
A	+100.00	+90.00	+10.00
B	+90.00	+81.00	+9.00
C	+81.00	+72.90	+8.10
D	+72.90	+65.61	+7.29
Total	+1000.00	+900.00	+100.00

Simple Deposit Multiplier

- To find out the total amount of deposits we must add up the change in deposits of the whole banking sector
- Technically this is the sum of geometric series (with quotient equal to minimum reserve requirements):

$$\Delta D = \frac{1}{r_D} \cdot \Delta R$$

Money Multiplier I.

- C/D = currency-checkable deposits ratio
- ER/D = excess reserves ratio
- RR = required reserves, $RR = r_D \cdot D$
- ER = excess reserves

$$R = r_D \cdot D + ER$$

$$MB = R + C = r_D \cdot D + ER + C$$

$$MB = r_D \cdot D + \frac{ER}{D} \cdot D + \frac{C}{D} \cdot D = \left(r_D + \frac{ER}{D} + \frac{C}{D} \right) \cdot D$$

Money Multiplier II.

$$MB = \left(r_D + \frac{ER}{D} + \frac{C}{D} \right) \cdot D$$

$$D = \frac{MB}{\left(r_D + \frac{ER}{D} + \frac{C}{D} \right)}$$

$$M = C + D = D + \frac{C}{D} \cdot D$$

$$M = \frac{1 + \frac{C}{D}}{r_D + \frac{ER}{D} + \frac{C}{D}} \cdot MB$$

What Influences Money Multiplier?

- Changes in required reserves ratio
 - Higher r_D means lower multiplier (cet. par.)
- Changes in currency/checkable deposits ratio
 - Higher C/D ratio means lower multiplier
- Changes in excess reserves ratio
 - Higher ER/D ratio means lower multiplier
- What determines these factors?
- Is the money multiplier stable?

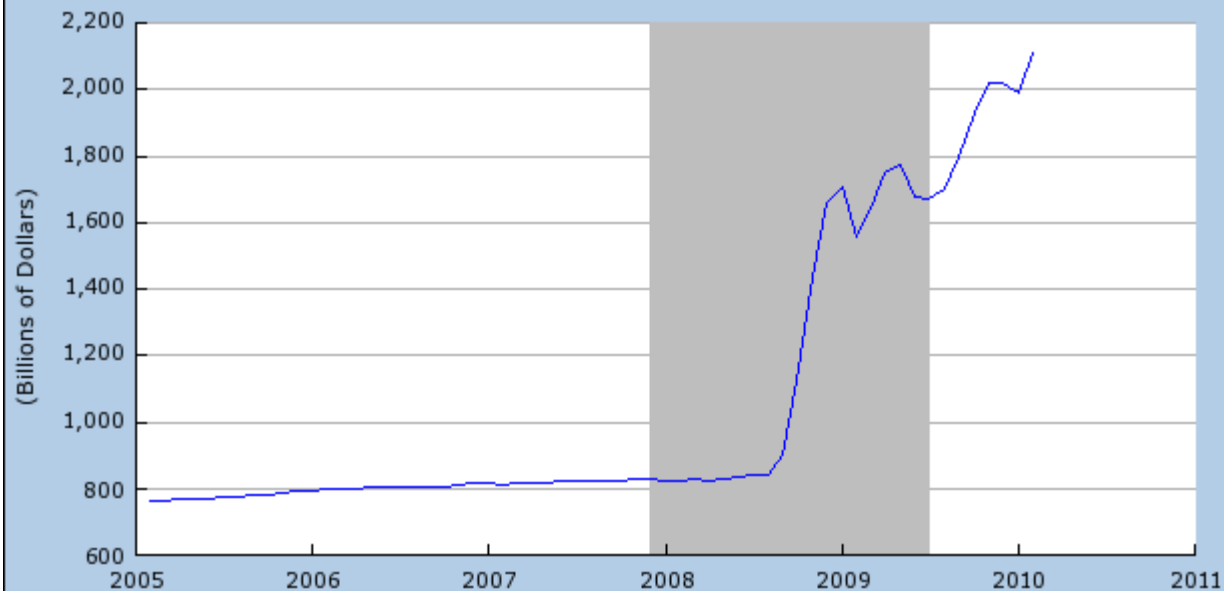
Endogenous Money?

- Do central banks really control money supply?
 - Their powers may be quite limited, because:
 - The selected instrument may not be able to influence the monetary base
 - The money multiplier is not under control of CEB
 - Other point of view: discussion about the stability of the velocity of money circulation
- ⇒ Alternative explanation for correlation of money and output – (post)keynesians strike back
- ⇒ Kaldor – money is endogenous, money supply tends to accommodate money demand

M1 Money Multiplier (MULT)
Source: Federal Reserve Bank of St. Louis



Board of Governors Monetary Base, Adjusted for Changes in Reserve Requirements (BOGAMBNS)
Source: Board of Governors of the Federal Reserve System



Shaded areas indicate US recessions.
2010 research.stlouisfed.org

Exchange Rate Regimes and Money Supply

- Exchange rate regimes
 - Floating
 - Managed floating
 - Pegged exchange rate
 - Currency board
- Pegged exchange rate: the CEB must intervene and prevent the currency from getting too far from the official central parity
 - Czech Republic 1991-1997
 - Countries in the ERM II
- Currency board: textbook version – a direct and stable link between the stock of foreign liquidity and domestic money supply
 - Hong Kong
 - More details: document “Introduction to HKMA” available at Moodle.

Sterilized Interventions

- Sterilization – attempt to neutralize monetary effects of the interventions
 - When the CEB increases FX reserves, it would be selling other assets at the same time
 - And vice versa
- Problems: efficiency and sustainability of sterilized interventions
- 1920s-1930s – sterilization attempted even under gold standard (UK, France)

References and additional reading

- M. F. Bryan: Island Money
- KOM: chapter 15
- CNB: main policy instruments -
http://www.cnb.cz/en/monetary_policy/instruments/
- Barro: Macroeconomics
- Dornbusch, Fischer, Starz: Macroeconomics
- Mishkin: Money, Banking and Financial Markets, 3rd Edition
- www.federalreserve.gov